

Problem 1 - Part (e)

Possibility of Further Model Size Reduction

Can you **further reduce the model size** beyond the smallest model obtained in parts **(b)**, **(c)**, or **(d)**, **without sacrificing significant classification performance**?

Your task is to:

1. **Analyze and compare** the results from previous parts: Which model had the smallest size? Which performed best?

The model that had the smallest size was the studentized version, and the best performer overall was the quantized version.

2. **Propose a strategy** that combines or enhances techniques learned so far.

A combination of the three techniques would lead to a model that benefits from all of the methodologies for compression. This should be done in the order of studentization to decrease the total model size, followed by pruning, then finally quantizing the whole thing.

4. **Implement** your proposed solution.

See jupyter notebook

5. **Evaluate** the resulting model using both:

- TFLite model size (in KB)
- Classification performance (accuracy and report)

The final result had a size of 3.70 KB. It performed with a minimal discernable decrease in F1 score in comparison to the best performance of the individual techniques.

6. **Justify your results:**

- If further size reduction is **not** possible without major loss of accuracy, explain why.

Further size reduction without significant accuracy loss is likely not possible as the current implementation already uses the three techniques to their theoretical maxima. This is because the studentization architecture is already minimal, the pruning sparsity is already aggressive, and that we already quantize to 8 bit integers.

Problem 2:

1. On Edge Impulse, the available learning blocks are supervised learning—Classification and Regression, unsupervised— Anomaly Detection (GMM) and (K-Means), as well as plain Anomaly Detection.
2. The types of layers available are the Input layer Dense (typical NN layer for processed data), 1D convolution/Pooling (learning of spatial information in the single dimension), 2D Convolution/Pooling (learning of spatial information in the second dimension), Reshape (converts 1D DSP blocks into multi dimension data for input into convolutional layers), Flatten (flattens multi-dimensional data into one dimension, usually for after convolution), and Dropout (reduces risk of overfitting by randomly cutting network connections).
3. The types of model compression available are Quantized (int8) and Unoptimized (float32).
4. For synthetic data generation in the free tier, there are four public blocks available on platform. These are Edge Impulse's Black Forest Labs Flux-Pro Synthetic Image Generator, DALL-E 3 Synthetic Image Generator, and Whisper Synthetic Voice Generator. Synthetic data is important in ML applications for a few reasons, first that real world data is often hard to source, some data is private and inaccessible, and rare events are often heavily imbalanced so there's not much to train on, to name a few reasons.